

Circuit Analysis (Lab # 6)

Experiment 6: VERIFICATION OF KIRCHHOFFS LAWS

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Grade and Signature:

CLO2	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
CLO3				
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objective

To verify Kirchhoff's current law and Kirchhoff's voltage law for the given Circuit.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	330ohm, 220ohm 1kohm	3

Description

KCL: The algebraic sum of the currents meeting at a node is equal to zero.

KVL: In any closed path / mesh, the algebraic sum of all the voltages is zero.

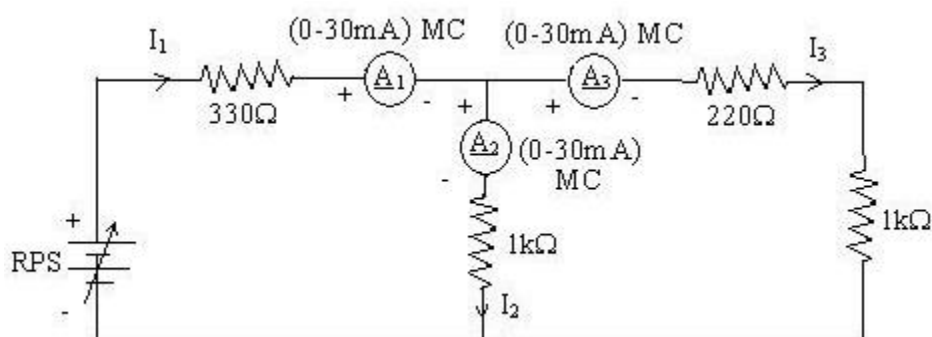
Precautions:

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.

Procedure for KCL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note down the corresponding readings using multimeter
4. Repeat the same for different voltages

Circuit - KCL



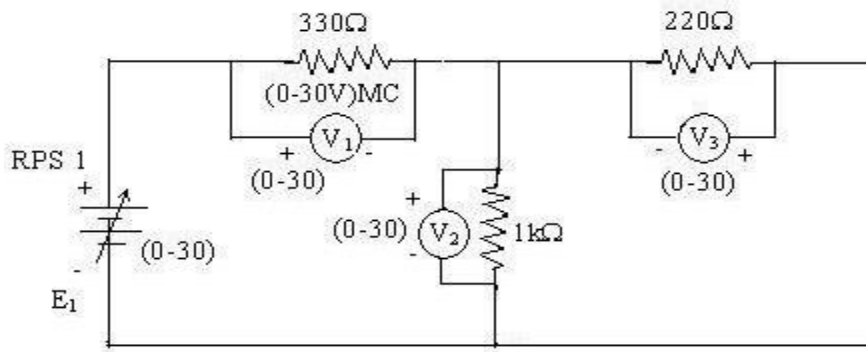
KCL - Practical Values:

Sl. No.	Voltage E Volts	Current			$I_1 = I_2 + I_3$ mA
		I_1 mA	I_2 mA	I_3 mA	
1	5	5.68	3.12	2.56	5.68
2	15	17.1	9.37	7.68	17.05
3	25	28.4	15.6	12.8	28.4

Procedure for KVL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note all the voltage reading
4. Repeat the same for different voltages

Circuit - KVL



KVL - Practical Values

Sl. No.	RPS	Voltage			$E1 = V1 + V2$
	E1	V1	V2	V3	
	V	V	V	V	
1	5	3.23	1.77	-1.77	5
2	10	6.47	3.53	-3.53	10
3	15	9.7	5.3	-5.3	15

Result:

Thus, Kirchhoff's voltage law and Kirchhoff's current law are verified practically by performing this experiment.

Post Lab Questions

1. Define Kirchhoff's current law.

The total current entering a node is equal to the total current leaving the node.

2. Define Kirchhoff's voltage law.

In any closed path/mesh, the algebraic sum of all the voltages is zero.

3. What are the applications of regulated power supply?

Provides stable DC Voltage to electronic devices`

Used in laboratories for testing experiments.

Used in communication systems.

4. What are the limitations of KCL?

It assumes ideal conditions

Not accurate at high frequencies(due to capacitances/inductances)

Difficult in complex circuits with many nodes.
